

PGM Lab

Illustrated user guide

Updated for version 5.3.0 - 8 October 2026

From dataset selection to the PGM, from statistical comparison to quantum-circuit execution on simulators or QPUs.

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<https://pgm-lab-sergioli.streamlit.app/>

1. Orientation and quick start

PGM Lab compares c-PGM, k-PGM, and r-PGM, three mathematically equivalent realizations of the Pretty Good Measurement classifier. The interface selects the least expensive exact backend without changing the decision rule.

Page map

PGM Lab		IT	EN
1	1 Dataset	Catalogue, upload, copies, and RAM	
2	2 Dimensions	Backend, memory, and circuit preview	
3	3 Comparison	Optional benchmark and full comparison	
4	4 Results	Metrics, samples, and complexity	
5	5 Quantum	Circuit, optimization, and providers	

Recommended path in 6 steps

1. Choose a public dataset or upload your own file.
2. Set the copy number c and, if needed, the RAM budget.
3. Check dimensions, expected backend, and feasibility.
4. Run the multi-seed PGM evaluation.
5. Read metrics, standard deviations, confusion matrix, and samples.
6. Only when needed, compare models or build and execute the circuit.

The flags in the upper-right corner switch the entire interface. Data, settings, and results are not recomputed when the language changes.

2. Public datasets and private upload

In section 1, open Upload your own dataset and drag the file into the dashed area. Once validated, it appears in the table and selector next to public datasets, marked with an upload icon.



1. Drag a CSV, TSV, TXT, or XLSX file (first Excel sheet).
2. Check the suggested target column and correct it if needed.
3. Confirm numeric features; likely identifiers are deselected.
4. Choose a readable name and inspect the preview and class counts.
5. Press Add to catalogue and select.

File requirements

Item	Meaning
Format	CSV, TSV/TXT, XLSX
Size	max 25 MB; 50,000 rows; 500 columns
Structure	first row with unique column names
Target	2-50 classes; at least 3 samples per class
Features	numeric; at least one varying feature
Samples	at least 10 after cleaning

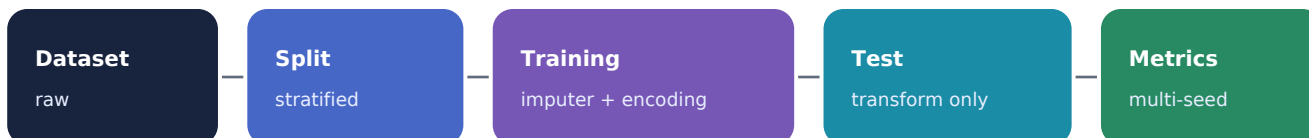
Rows without a target are removed. Missing feature values are imputed with the median learned only from each seed's training set. Text, empty, and constant columns are reported clearly.

A private file stays in session memory, never enters the repository, and does not use the shared cache for public datasets. Remove private dataset also clears results that depend on that file.

3. Scientific settings and dimension checks

By default, the PGM uses every selected feature. PCA is never automatic: it is applied only when the user enables Manually request PCA reduction.

LEAKAGE-SAFE WORKFLOW



The test set never selects imputation, PCA, scaling, encoding, or t.

Item	Meaning
Copy number c	kernel power and symmetric-map order
Test fraction	only values compatible with a stratified split
Seeds	reproducible splits; default: 10 consecutive seeds
Class prior	uniform or empirical class frequencies
Spectral threshold	controls numerical rank consistently
RAM budget	safety limit; it does not increase server RAM

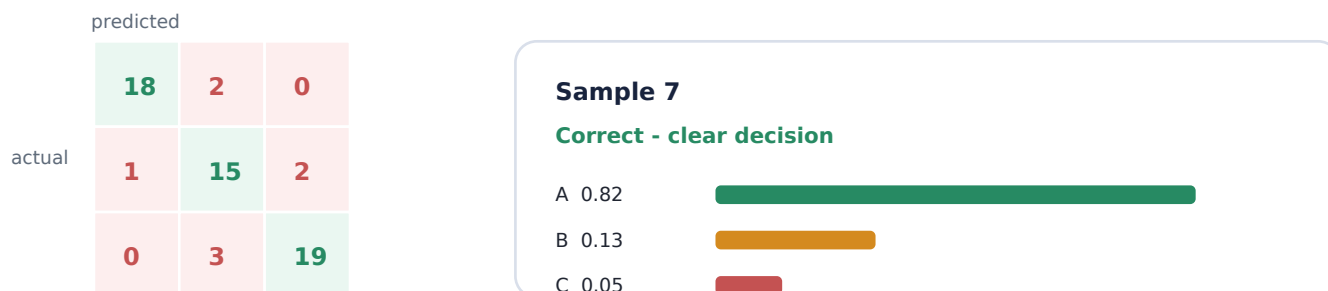
On each training set, the app compares amplitude encoding with stereographic encoding at $t = 0.1, 0.2, 0.5, 1, 2$. The test set never participates in this choice. The winning encoding is reported.

When the circuit is too large, classical classification may remain available. The exact circuit is materialized up to 9 qubits; automatic full synthesis is prudently limited to 7 qubits.

4. Running the PGM and interpreting results

Press Run the multi-seed evaluation. The main summary uses mean and standard deviation across all splits; one reference seed is used only for diagnostics, sample details, and circuit construction.

READING EXAMPLE



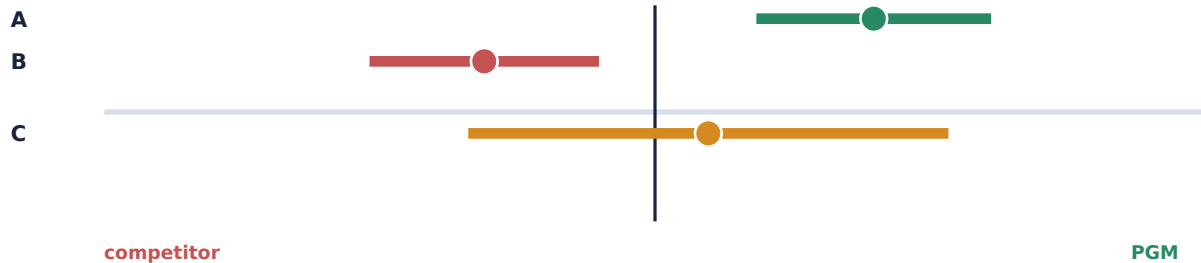
Item	Meaning
Balanced accuracy	mean class recall; useful for imbalanced classes
Accuracy	overall fraction of correct predictions
Macro precision	precision computed per class and then averaged
Macro recall	mean sensitivity across classes
Macro F1	macro harmonic mean of precision and recall
Kappa / MCC	chance-corrected agreement / global correlation
ROC-AUC	ranking quality; may be unavailable for some models

In Result for each sample, filter correct or incorrect cases. Confidence is the probability of the selected class; Margin is the gap to the runner-up. Circuit-theory deviation should remain near numerical tolerance.

5. Comparison with other classifiers

This section is optional and does not run with the PGM button. Enable it, choose a competitor, and start the selected-dataset comparison. Both models use the same seeds and outer splits.

BALANCED-ACCURACY DIFFERENCE



The segment is the 95% interval; the point is the mean difference.

- Competitor tuning uses training data only.
- Balanced-accuracy differences are paired seed by seed.
- Green: confirmed PGM win; red: confirmed competitor win.
- Yellow: visible direction, but the 95% interval includes zero.
- Gray: exact numerical equality on every seed.
- Full comparison repeats the analysis on available binary datasets.

Yellow does not mean a tie: it shows a mean advantage whose uncertainty does not yet support a 95% conclusion.

6. Quantum circuit and certified optimization

After training, the app constructs the reduced POVM and the Naimark dilation U_{PGM} . The sys register contains the encoded test state; measuring the out register returns the class index.

NAIMARK DILATION



Optimized only if $||U_{\text{ref}} - U_{\text{opt}}|| \leq \text{tolerance}$

000 -> class 1

- The bitstring-to-class table shows mappings such as 000 -> class 1.
- Circuit probabilities are compared with theoretical probabilities.
- Full decomposition runs outside the Streamlit process.
- Optimization levels 1, 2, and 3 are compared.
- A circuit is accepted only after numerical equivalence certification.
- For huge diagrams, gate count, depth, and QPY download remain available.

Optimized does not mean approximated: the app retains only circuits that pass equivalence certification.

7. Simulators, cloud services, and real QPUs



Mode	Availability
Local	ideal PGM sampler; Qiskit Aer
LRZ / MQSS	authorized resources discovered dynamically
IBM Quantum	simulators or QPUs available to the account
IonQ Cloud	IonQ simulators and QPUs
Amazon Braket	managed simulators and supported QPUs

1. Choose a simulator or QPU and configure shots and options.
2. Enter credentials only in the protected field or secrets.
3. Discover devices; the list depends on account rights and online status.
4. Run preflight and check qubits, gates, and potential cost.
5. Explicitly confirm submission, then inspect status and result.

Caution: cloud simulator or hardware jobs may incur costs and queues. PGM Lab embeds no tokens, cannot guarantee device availability, and submits nothing before final confirmation.

8. Downloads, privacy, and good practice

PRIVATE SESSION

- uploaded file
- results and circuit
- no shared cache

PUBLIC RESOURCES

- OpenML/UCI catalogue
- download cache
- PDF guides

NO

Format	Contents
CSV	predictions and detailed classifications
SVG / TXT	logical diagram or readable decomposition
QPY	serialized Qiskit circuit
NPZ	U_PGM, effects, classes, and encoding metadata

- Never place tokens in code, CSV files, or the repository.
- Immediately revoke a credential published by mistake.
- Check dataset permissions before uploading sensitive data.
- Scientific results depend on splits, seeds, priors, and preprocessing.
- Store the app version and parameters alongside exports.

9. Quick troubleshooting

Problem	Recommended action
File is not read	check format, headers, delimiter, and size limits
No features	select varying numeric columns; exclude text and IDs
Invalid split	use an offered fraction or add samples to rare classes
Computation blocked	reduce c, raise budget only with real RAM, or request PCA
Circuit not built	classification stays valid; reduce dimensions or copies
Provider absent	check packages, credentials, region, and authorization
Connection error	inspect Streamlit logs and restart the service

For a controlled first run, use Iris, $c=2$, and 3 seeds, then move to 10 seeds. With a private file, first inspect class distribution and one run; only then enable full comparison or remote hardware.

This guide is versioned with the app. If APP_VERSION changes, the tests require guide regeneration before release.